PRANAVI JAIN

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SUMMARY

Quantum Software Engineer with experience developing quantum circuit and algorithm benchmarking frameworks, error mitigation pipelines, and hybrid quantum-classical simulations across multiple simulators and cloud backends. Background in nanofabrication, semiconductor devices, and large-scale enterprise systems. IEEE-published researcher with strong Python, Java, and distributed systems experience.

EDUCATION

University of Southern California (USC) — Los Angeles, CA, USA
M.S. Quantum Information Science | GPA – 3.57/4 — August 2023 – May 2025

Thapar Institute of Engineering and Technology — Patiala, Punjab, India
B.E. Electronics and Computers | CGPA – 9.03/10 — July 2017 – June 2021

WORK EXPERIENCE

NeoQuortex — College Park, MD, USA
Quantum Software Engineer — June 2025 – *Present*

- Developed PennyLane-based Python framework for characterizing parameterized quantum circuits and ansatz with six different metrics, including expressibility and entanglement capability.
- Designed a benchmarking pipeline comparing 6+ parameterized circuits across 3+ algorithms for H2 and LiH molecules, producing comparative performance datasets used as guide for internal ansatz selection for different algorithms.
- Conducted literature-driven research in quantum chemistry, hardware-efficient ansatz, circuit knitting, and circuit optimization; translated results into production-ready prototypes.

USC Nanostructures Materials and Devices Laboratory — Los Angeles, CA, USA
Graduate Researcher — May 2024 – December 2024

- Fabricated GaAs waveguides using RIE for Mesa-Top Quantum Dots; executed DOE-driven experimental workflows.
- Operated cleanroom tools including plasma etchers, profilometers, optical microscopy, wet benches, and fumehoods.

Amdocs — Gurgaon, NCR, India
Software Developer — July 2021 – July 2023

- Delivered Java-based backend services for telecom OMS/CRM platforms supporting high-revenue customer deployments.
- Led PoCs directly with Singtel clients and supported production releases, including the release for Singtel RED.
- Built SOAP/XML APIs and database integrations for Singtel's digital platforms.
- Handled documentation and conducted sessions for knowledge transfer among the team.

Saathi Global Education Network — Remote, India
Web Development Intern — January 2021 – June 2021

- Led a 3-person team building Django-based CMS platform with REST APIs and Neo4j backend.

PUBLICATIONS

IEEE ICIBT 2022 – **Post Quantum Digital Signature Change in IOTA to Reduce Latency in Internet of Vehicles (IoV) Environments**

ACADEMIC PROJECTS

- **Towards Benchmarking Quantum Computers: Implementing Grover's Algorithms** – Running Grover's algorithm on various available quantum simulators and hardware: IBM (via Qiskit), IonQ, IQM, and Rigetti (via AWS Braket) to benchmark quantum computers.
- **A Review of Zero-Noise Extrapolation (ZNE) Technique for Quantum Error Mitigation** – A report reviewing fundamentals of the ZNE method and its various modifications, including an implementation of ZNE using Qiskit with mitiq Python package.
- **Examining Two-Qubit Entanglement Dynamics on IBM Quantum Systems** – An experimental study of entanglement dynamics on IBM quantum processors by simulating the Nakajima-Zwanzig and Time-Convolutionless (TCL) master equations.
- **Hybrid Quantum Networks for End-to-End Qubit State Transfer** – A theoretical report on integrating Fluxonium qubits with piezoelectric membranes for optical transduction of quantum states, enabling seamless transfer of information without loss.

SKILLS

- **Programming:** Python, C/C++, Java, ARM Assembly, MySQL
- **Quantum:** PennyLane, Qiskit, AWS Braket, OpenQASM, PySCF, OpenFermion
- **Systems:** Linux/Unix, Git/GitHub, Perforce, MATLAB
- **Web/APIs:** Django, XML, REST, SOAP, Postman
- **Hardware:** Arduino, Raspberry Pi, Nanofabrication – Profilometry, Scribe, RIE